

United International University

CSE 1110 (Section D) – Final Examination

Time: 60 Minutes, Full Marks: 20

Marks Obtained

Name:

ID:

There are 2 questions. **Answer all questions in Code::Blocks IDE.** You can do rough works in this sheet.

Problem 1 (Marks: 10)

A popular theme park determines ticket prices based on the visitor's category code (1 for Child, 2 for Adult, 3 for Senior, 4 for Veteran) and whether the visit is on a weekend.

Write a C program that prompts the user for their category code and a weekend status indicator (1 for Weekend, 0 for Weekday). The program should calculate the final ticket price based on these rules:

- **Child (Category 1):** Tickets are a flat rate of \$20, regardless of the day.
- **Adult (Category 2):** The standard base price is \$40. If it is a weekend (1), add a weekend surcharge of \$15. Otherwise, no surcharge is added.
- **Senior (Category 3):** The standard base price is \$30. If it is a weekend (1), apply a small discount of \$3. If it is a weekday (0), apply a larger discount of \$8.
- **Veteran (Category 4):** Veterans receive a flat rate of \$25, regardless of the day.
- **Any other category:** Print an invalid category error and set the ticket price to \$0.

You have to calculate the ticket price according to the above rules, and print the final price.

Sample input (Category, Weekend)	Sample output
Category: 2 Weekend: 1	Ticket price: \$55
Category: 3 Weekend: 0	Ticket price: \$22
Category: 5 Weekend: 0	Invalid category! Ticket price: \$0

Problem 2 (Marks: 10)

Write a C program that will take as input an integer n , and count the total number of “high digits” in the number. A high digit is any digit greater than or equal to 5.

Sample input	Sample output
27591	3 <i>(Explanation: 7, 5 and 9 are the high digits)</i>
132042	0 <i>(Explanation: No digits are high)</i>
8688	4 <i>(Explanation: All digits are high)</i>

United International University

CSE 1110 (Section D) – Final Examination

Time: 60 Minutes, Full Marks: 20

Marks Obtained

Name:

ID:

There are 2 questions. **Answer all questions in Code::Blocks IDE.** You can do rough works in this sheet.

Problem 1 (Marks: 10)

A company is calculating year-end bonuses for its employees based on their Department Code (1 for Sales, 2 for Tech, 3 for HR, 4 for Admin) and their total years of experience.

Write a C program that takes the Department Code (integer), the employee's current annual salary (float), and their years of experience (integer) as inputs. Compute the bonus amount using the following logic:

- **Department 1 (Sales):** The bonus is entirely percentage-based. If the employee has more than 5 years of experience, the bonus is 15% of their annual salary. Otherwise, the bonus is 10% of their annual salary.
- **Department 2 (Tech):** The bonus is a mix of percentage and experience. If the employee has more than 3 years of experience, the bonus is 12% of their salary. Otherwise, the bonus is 8% of their salary.
- **Department 3 (HR):** Employees receive a flat bonus of 7% of their annual salary, regardless of experience.
- **Department 4 (Admin):** Employees receive a flat bonus of \$2,500, regardless of salary or experience.
- **Any other department:** The code is invalid. Set the bonus to \$0 and print an error message.

Perform the calculations as per the above conditions and print the final calculated bonus amount.

Sample input (Department, Salary, Experience)	Sample output
Department: 1 Salary: 60000.00 Experience: 6	Bonus Amount: \$9000.00
Department: 2 Salary: 80000.00 Experience: 2	Bonus Amount: \$6400.00
Department: 4 Salary: 50000.00 Experience: 10	Bonus Amount: \$2500.00

Problem 2 (Marks: 10)

Write a C program that will take as input an integer n , and count how many times the last digit (i.e. the digit in the one's place) appears in the number. *The last digit itself will always count as the first match.*

Sample input	Sample output
3523	2 <i>(Explanation: Last digit is 3, appears twice)</i>
777	3 <i>(Explanation: The whole 3 digit number consists of 7 only)</i>
1482	1 <i>(Explanation: Last digit is 2, appears only once)</i>